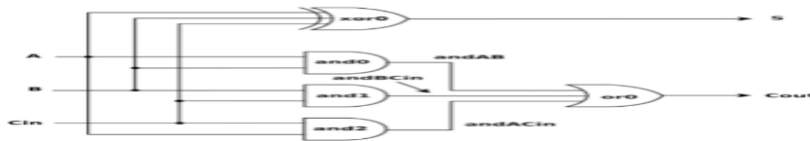
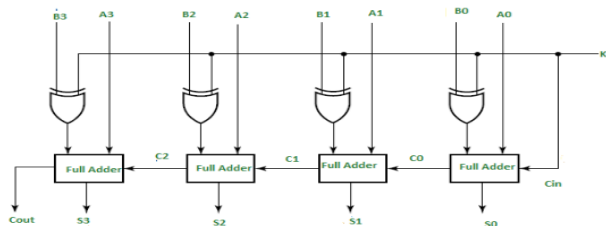


Suggestions: FOR Group A: 03.06.2026

Q1. Write a verilog program to design, simulate and test the following circuit.



Q2. Using Module instantiation, write down a verilog code to design, simulate and test the following circuit.

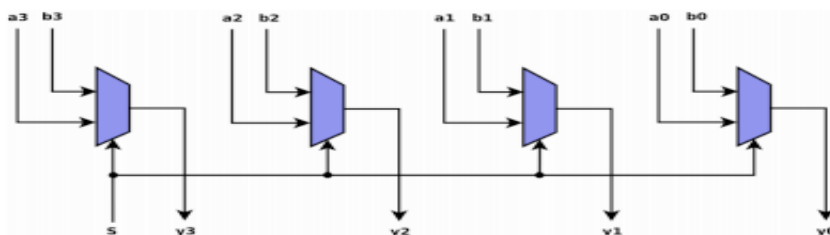


Q3. Write a verilog program to design, simulate and test the circuit of a Full Adder.

Q4. Design a 4-bit Incrementor /Decrementor composite circuit using Full adder circuit

Q5. Write a Verilog program to design, simulate and test a circuit of a 2x1 MUX using TERNARY operator.

Q6. Using Module instantiation of part, A, write down a Verilog code to design, simulate and test the following circuit. Maintain the port names as shown in the diagram.



Q7. Write a Verilog program to design, simulate and test the circuit of a Half Adder.

Q8. Write a Verilog program to design, simulate and test the circuit implemented using Decoder for the following three Boolean functions:

$$F_1 = \Sigma (1, 4, 7)$$

$$F_2 = \Sigma (0, 3)$$

$$F_3 = \Sigma (0, 2, 6, 4, 7)$$

Q9. Write a Verilog program to design, simulate and test the circuit of a Full Adder.

Q10. Write a Verilog program to design, simulate and test the circuit of a 4-bit Adder/Subtractor composite circuit using module instantiation of Full Adder.

Q11. Write a Verilog program to design, simulate and test the circuit of a Half Adder.

Q12. Write a Verilog program to design, simulate and test the circuit of a Full Adder using module instantiate of Half Adder.

Q13. Write a Verilog program to design, simulate and test the circuit implemented using Decoder for the following three Boolean functions:

$$F_1 = \Sigma (2, 4, 7)$$

$$F_2 = \Sigma (0, 3)$$

$$F_3 = \Sigma (0, 2, 3, 4, 7)$$

Q14. Write a Verilog program to design, simulate and test the circuit of a 4-bit adder/subtractor

composite circuit.

Q15. Write a Verilog program to design, simulate and test the circuit to Implement the following function using basic gates. $F = \sum m(1,3,7,8,12,13,14,15)$

Q16. Write a Verilog program to design, simulate and test the circuit of a Full Subtractor.

Q17. Write a Verilog program to design, simulate and test the circuit to Implement the following function using basic gates.

$F = \sum m(0,2,4,7,8,9,11,12,13,14,15)$.

Q18. Write a Verilog program to design, simulate and test the circuit of a Half Subtractor.

Q19. Write a Verilog program to design, simulate, and test the circuit of a 3:8 decoder.

Q20. Write a Verilog program to design, simulate, and test the circuit of a Full Adder using 3:8 a decoder

Q21. Write a Verilog program to design, simulate and test the circuit of a Half Adder.

Q22. Write a Verilog program to design, simulate and test the circuit implemented using Decoder for the following three Boolean functions:

$$F_1 = \sum (1, 4, 7)$$

$$F_2 = \sum (0, 3)$$

$$F_3 = \sum (0, 2, 6, 4, 7)$$

Q23. Write a Verilog program to design, simulate, and test the circuit of a 4:2 priority encoder.

Q24. Write a Verilog program to design, simulate, and test the circuit to Implement the following function using basic gates. $F = \sum m(0,2,4,7,8,9,11,12,13,14,15)$.

Q25. Write a Verilog program to design, simulate, and test a circuit of a 2:1 MUX using the TERNARY operator.

Q26. Write a Verilog program to design, simulate, and test the circuit of a 4:1 Mux using module instantiate of 2:1 MUX

Q27. Write a Verilog program to design, simulate, and test the circuit to Implement the following function using basic gates.

$F = \sum m(1,3,7,8,12,13,14,15)$

Q28. Write a Verilog program to design, simulate, and test the circuit of a Full Adder using module instantiate of Half Adder.

Q29. Write a Verilog program to design, simulate and test the circuit to Implement the following function using basic gates.

$F = \sum m(0,2,4,7,8,9,11,12,13,14,15)$.

Q30. Write a Verilog program to design, simulate and test the circuit of a Half Subtractor.